GEOMETRY	PLANES	PIXEL	BUS
960 x 672 px, 672 lines 120 B/line = 60 B CS1 + 60 B CS2	2 x 80 640 B, 1 bpp, MSB first x=0 is bit 7. Both always sent	table = (red<<1) | bw 0 paper 1 ink 2 red 3 unused 4 VCOM	regs + OTP bitbang ~50 kHz planes HW SPI 4 MHz, CS per 60 B

1 OTP probe ONCE PER POWER CYCLE, THEN CACHED

#	INDEX	BYTES	MEANING
1	—	—	VCC_EN on 800 ms; RST hi 50 / lo 100 / hi 400
2	0xA1	00	unlock step 1
3	0xA7	80	OTP strobe
4	0xA3	00, 40	start address
5	0xA8	—	read opener, index only, no data phase
6	—	1 + 88 B	SDA to input; dummy byte, then the block
7	—	—	OTP[00] must read **A5**, else the probe failed

OTP OFFSETS USED

0B → 0x60 TCON
0C–0F → 0x90 DRFW
10 → 0x01 DCTL
11 → 0x02
12 → 0x03 osc
13–15 → 0x12 RAM_RW
16–1B → 0x13 DUW
1C → 0x61 STV
1D → 0xD8
1E → 0xD6

20–3F → DC/DC ramp, 4 stages x 8 B

RAMP STAGE, 8 B

[0] bit7 = mode, bits 6:0 = repeat 1..40
ramped [1][2] ph init · [3][4] step · [5][6] 0x09 pre/post · [7] delay
plain [1][2] 0x09 a/b · [3][4] delays
delay byte: bit7 → (v&7F) ms, else x 10 us
Repeat runs verbatim: FF asks 127 rounds and hangs BUSY, so all four stages are checked first.

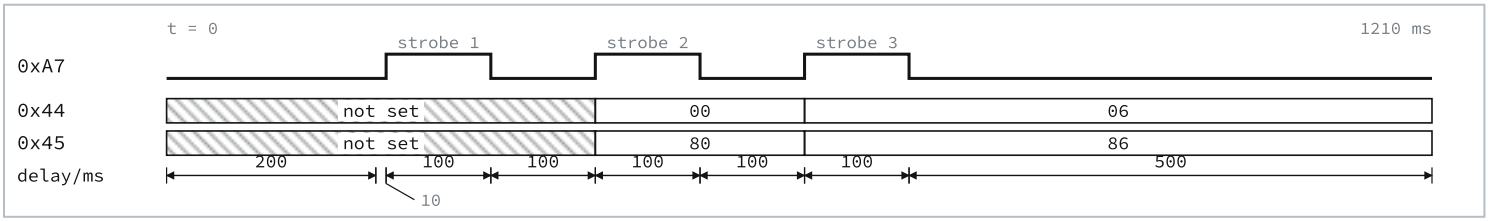
2 Initialisation, one image EXECUTION ORDER

#	INDEX	BYTES	MEANING	DELAY / MS
POWER AND RESET				
1	—	—	VCC_EN on, after the 1500 ms minimum supply-off time	800
2	—	—	RST high, then low, then high	50/100/400
FRAME DATA				
3	0x13	OTP[16..1B]	DUW. [0] x start · [1] x end = **3B** = 60 B per COG · [2..5] y window, verbatim	—
4	0x90	OTP[0C..0F]	DRFW. [1] x end = **3B** · [0][2][3] forwarded verbatim	—
5	0x12	OTP[13..15]	RAM_RW pointer, 3 B verbatim	—
6	0x10	80 640 B	bw plane. 672 x (60 B CS1 low + 60 B CS2 low)	—
7	0x12	OTP[13..15]	RAM_RW again — re-sent before every plane	—
8	0x11	80 640 B	red plane. Always sent; zeros if the frame has no red	660
REGISTER SEQUENCE — 1210 MS				
9	0x05	7D	opens the sequence	200
10	0x05	00	—	10
11	0xC2	3F	panel setting, literal	—
12	0xD8	OTP[1D]	panel setting from the fuse block	—
13	0xD6	OTP[1E]	panel setting from the fuse block	—
14	0xA7	10	strobe 1 high	100
15	0xA7	00	strobe 1 low	100
16	0x03	00, OTP[12]	oscillator, injected right after strobe 1 falls	—
17	0x44	00	waveform source pointer	—
18	0x45	80	bank, driver default	—
19	0xA7	10	strobe 2 high	100
20	0xA7	00	strobe 2 low	100
21	0x44	06	waveform source pointer	—
22	0x45	86	bank. **86** = 9.7 in, 8C = 12 in	—
23	0xA7	10	strobe 3 high	100
24	0xA7	00	strobe 3 low — commits the OTP waveform	500
25	0x60	OTP[0B]	TCON, written before STV	—
26	0x61	OTP[1C]	STV	—
27	0x01	OTP[10]	DCTL	—
28	0x02	OTP[11]	closes the sequence	—
WAVEFORM, BOOSTER, TRIGGER				
29	0xB0	180 B	short waveform, fast refresh only. **The only window for it**	—
30	0x09 / 0x51	OTP[20..3F]	DC/DC soft start, four stage records	100
31	—	—	wait for BUSY high, ceiling 10 s, then proceed anyway	—
32	0x15	3C	trigger, **once** — the table sets the duration	—
33	—	—	BUSY low <2 s, high <180 s; watch 1 s for a second pass	—

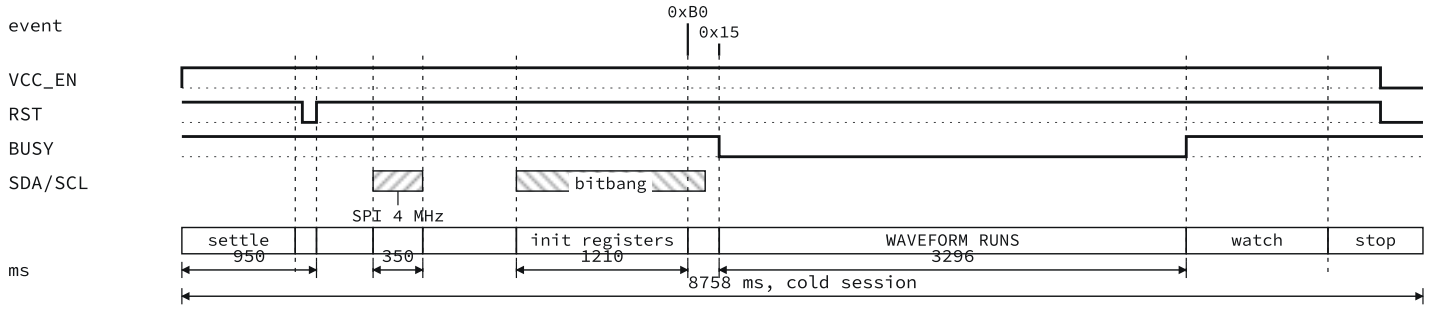
One window only. Earlier the strobes overwrite 0xB0; later the booster is up. Rewriting it inside an open session does not re-latch — it drops back to the OTP waveform.

3 Shutdown. 0x09←7F, 20 ms · 0x05←7D · 0x09←00, 200 ms · DC and both CS low, 150 ms · RST low, VCC_EN released, 300 ms · CS and RST high.

4 The sequence in time STEPS 9–28



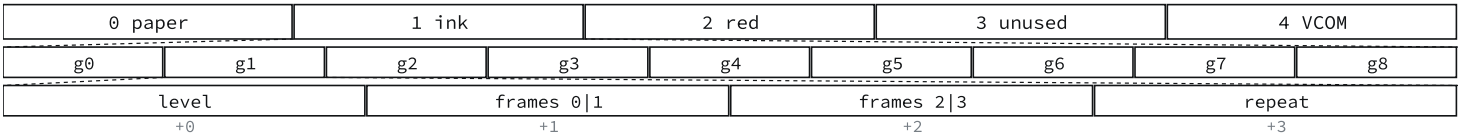
5 Refresh timing FAST REFRESH, COLD SESSION



One trigger, and BUSY stays low for the whole 3296 ms the table asks for. A slow refresh is the same picture without step 29, with the low period at about 51 s. A kept-open session re-enters at the plane write and ends at the trigger.

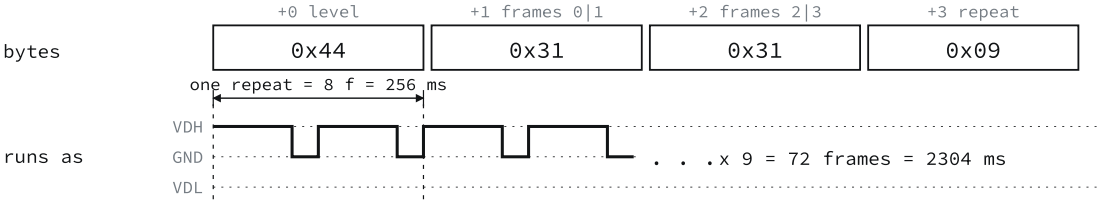
6 Waveform table at 0xB0 180 BYTES

0xB0 payload, 180 B = 5 tables x 9 groups x 4 bytes



level byte = 2 bits per phase, phase 0 in bits 7:6. 0 = GND, 1 = VDH, 2 = VDL, 3 = VDHR (reduced positive, red only).
offset of table t, group g, byte n = (t * 9 + g) * 4 + n

WORKED EXAMPLE table 1 (ink), group 1



One frame = 32.0 ms. Three at the rail, one resting at ground: a long run at VDH drags the slower red pigment along behind the black.

7 Complete table ALL 180 BYTES

TABLE	GROUP	BYTES	PHASE LEVELS	FRAMES	REP	DURATION	ROLE
0 paper	0	66 33 33 02	VDH VDL VDH VDL	3 3 3 3	2	24 f / 768 ms	balanced shake
0 paper	1	88 31 31 09	VDL GND VDL GND	3 1 3 1	9	72 f / 2304 ms	chopped drive to paper
0 paper	2–8	00 10 00 01	GND x 4	1 0 0 0	1	1 f / 32 ms	placeholder
1, 2, 3	0	99 33 33 02	VDL VDH VDL VDH	3 3 3 3	2	24 f / 768 ms	shake, opposite phase
1, 2, 3	1	44 31 31 09	VDH GND VDH GND	3 1 3 1	9	72 f / 2304 ms	chopped drive to ink
1, 2, 3	2–8	00 10 00 01	GND x 4	1 0 0 0	1	1 f / 32 ms	placeholder
4 VCOM	0	00 33 33 02	GND x 4	3 3 3 3	2	24 f / 768 ms	ground, group timing
4 VCOM	1	00 31 31 09	GND x 4	3 1 3 1	9	72 f / 2304 ms	ground, group timing
4 VCOM	2–8	00 10 00 01	GND x 4	1 0 0 0	1	1 f / 32 ms	ground, group timing

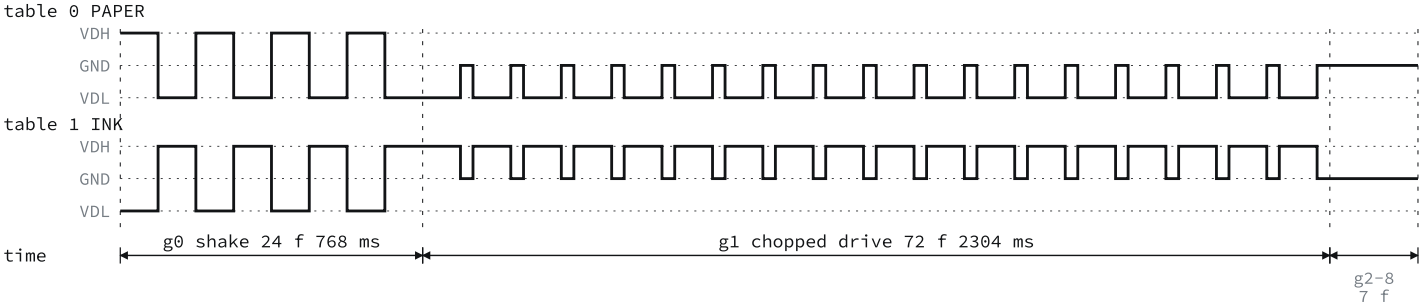
103 frames, 3296 ms against about 51 s for the OTP waveform. The shake is balanced — as much VDH as VDL — so it adds no net charge; without that the paper goes pink within a few updates. Groups 2–8 are 1-frame placeholders: a group left at zero runs about four seconds.

8 When the frame carries red GROUPS 2–8 REWRITTEN

CASE	TABLE	GROUP	BYTES	LEVELS	PURPOSE
red drawn	0 paper	2–8	88 66 66 01	VDL GND VDL GND	held at its own target, not parked
red drawn	1, 3 ink	2–8	44 66 66 01	VDH GND VDH GND	held at its own target, not parked
red drawn	2 red	2	88 66 66 01	VDL GND VDL GND	erase to white first
red drawn	2 red	3–8	FF 66 66 01	VDHR x 4	hold the reduced positive rail
clearing	2 red	2–8	88 66 66 01	VDL GND VDL GND	all seven erase: red to remove, none to make
both	4 VCOM	2–8	00 66 66 01	GND x 4	ground, with the group timing

Frame byte 66 = 6 frames per phase = 192 ms, so a rewritten group is 24 frames and the table grows to **264 frames, 8.45 s**. Groups 0 and 1 are never rewritten. Red is driven on **VDHR**, not VDH: both pigments rise on a positive field and differ only in threshold, so on the full rail black comes up with the red and the pull that removes black removes red too.

9 The whole waveform, frame by frame



All 103 frames to scale, 3296 ms. VCOM sits at GND throughout but carries the same frame counts: a group ends when its shortest table does.

GEOMETRY	PLANES	PIXEL	BUS
960 x 672 px, 672 lines 120 B/line = 60 B CS1 + 60 B CS2	2 x 80 640 B, 1 bpp, MSB first x=0 is bit 7. Both always sent	table = (red<<1) | bw 0 paper 1 ink 2 red 3 unused 4 VCOM	regs + OTP bitbang ~50 kHz planes HW SPI 4 MHz, CS per 60 B

1 OTP probe ONCE PER POWER CYCLE, THEN CACHED

#	INDEX	BYTES	MEANING
1	—	—	VCC_EN on 800 ms; RST hi 50 / lo 100 / hi 400
2	0x70	71, 36	booster config, to **both** COGs. Without it the panel draws ~2 A refreshing
3	0xB9	—	read opener, index only, **master COG only**
4	—	1 + 88 B	SDA to input; dummy byte, then the block
5	—	—	identified by OTP[16]=3B, OTP[0D]=3B, OTP[28] bit 7. No A5 header
6	—	—	remapped, then OTP[00] forced to A5

OTP OFFSETS, AFTER THE REMAP

0B → 0x60 TCON
0C–0F → 0x90 DRFW
10 → 0x01 DCTL
11 → 0x02
13–15 → 0x12 RAM_RW
16–1B → 0x13 DUW
1C → 0x61 STV
1D → 0xD8 · 1E → 0xD6

20–3F → DC/DC ramp, 4 stages x 8 B

THE REMAP

Raw block: 12–1D sit **one byte lower**, ramp at **28–47**. Fields 12..1D shift up one, the ramp block down eight. The byte landing in both RAM_RW[0] and the oscillator slot is harmless.

RAMP STAGE, 8 B

[0] bit7 = mode, bits 6:0 = repeat 1..40
ramped [1][2] ph init · [3][4] step · [5][6] 0x09 pre/post · [7] delay
plain [1][2] 0x09 a/b · [3][4] delays
delay byte: bit7 → ms, else x 10 us. FF hangs BUSY.

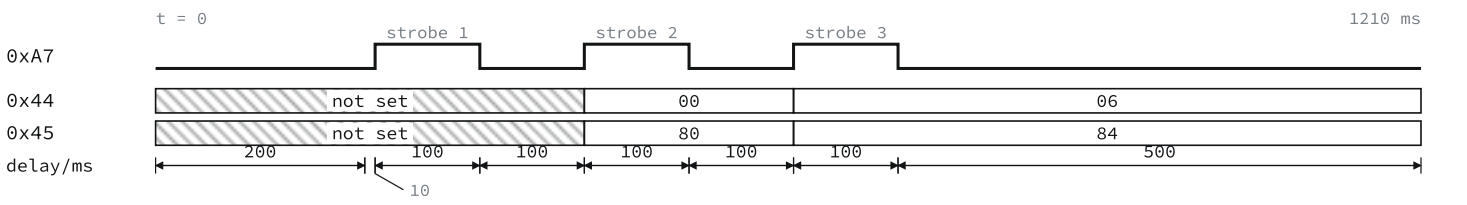
2 Initialisation, one image EXECUTION ORDER

#	INDEX	BYTES	MEANING	DELAY / MS
POWER AND RESET				
1	—	—	VCC_EN on, after the 1500 ms minimum supply-off time	800
2	—	—	RST high, then low, then high	50/100/400
FRAME DATA				
3	0x01	OTP[10]	DCTL. **Sent here on this variant**, ahead of the windows	—
4	0x13	OTP[16..1B]	DUW. [0] x start · [1] x end = **3B** = 60 B per COG · [2..5] y window, verbatim	—
5	0x90	OTP[0C..0F]	DRFW. [1] x end = **3B** · [0][2][3] forwarded verbatim	—
6	0x12	OTP[13..15]	RAM_RW pointer, 3 B verbatim	—
7	0x10	80 640 B	bw plane. 672 x (60 B CS1 low + 60 B CS2 low)	—
8	0x12	OTP[13..15]	RAM_RW again — re-sent before every plane	—
9	0x11	80 640 B	red plane. Always sent; zeros if the frame has no red	660
REGISTER SEQUENCE — 1210 MS				
10	0x05	7D	opens the sequence	200
11	0x05	00	—	10
—	0xC2	—	not written on this variant	—
12	0xD8	OTP[1D]	panel setting from the fuse block	—
13	0xD6	OTP[1E]	panel setting from the fuse block	—
14	0xA7	10	strobe 1 high	100
15	0xA7	00	strobe 1 low. **No oscillator write follows** — 0x03 is never sent	100
16	0x44	00	waveform source pointer	—
17	0x45	80	bank, driver default	—
18	0xA7	10	strobe 2 high	100
19	0xA7	00	strobe 2 low	100
20	0x44	06	waveform source pointer	—
21	0x45	84	bank. **84** on this variant	—
22	0xA7	10	strobe 3 high	100
23	0xA7	00	strobe 3 low — commits the OTP waveform	500
24	0x60	OTP[0B]	TCON, written before STV	—
25	0x61	OTP[1C]	STV	—
—	0x01	—	not written here — it went out at step 5	—
26	0x02	OTP[11]	closes the sequence	—
WAVEFORM, BOOSTER, TRIGGER				
27	0xB0	400 B	short waveform, fast refresh only. **The only window for it**	—
28	0x09 / 0x51	OTP[20..3F]	DC/DC soft start, four stage records	100
29	—	—	wait for BUSY high. Usually **still ramping** here; triggering early is the sits-still-then-draws fault	—
30	0x15	3C	pull 1	~1070
31	0x15	3C	pull 2. No plane data, no register touched	~1070
32	0x15	3C	pull 3	~1070
33	—	—	watch 1 s for a second pass	—

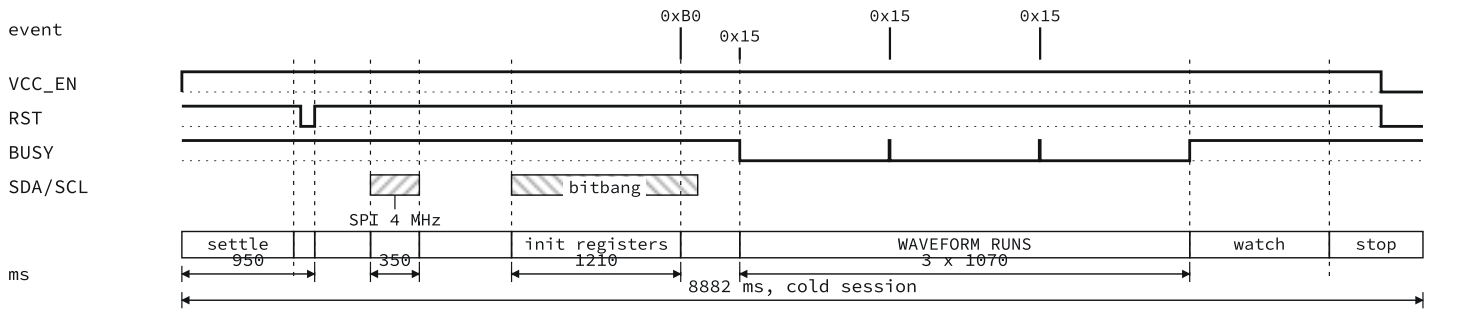
One window only. Earlier the strobes overwrite 0xB0; later the booster is up. Rewriting it inside an open session does not re-latch — it drops back to the OTP waveform.

3 Shutdown. 0x09←7F, 20 ms · 0x05←7D · 0x09←00, 200 ms · DC and both CS low, 150 ms · RST low, VCC_EN released, 300 ms · CS and RST high.

4 The sequence in time STEPS 10–26



5 Refresh timing FAST REFRESH, COLD SESSION



Three separate BUSY low periods of about 1070 ms. Everything up to the trigger is loaded once; a pull is a bare `0x15`. A slow refresh is the same picture without step 27, one pull, and the low period at about 34 s.

6 Waveform table at 0xB0 400 BYTES

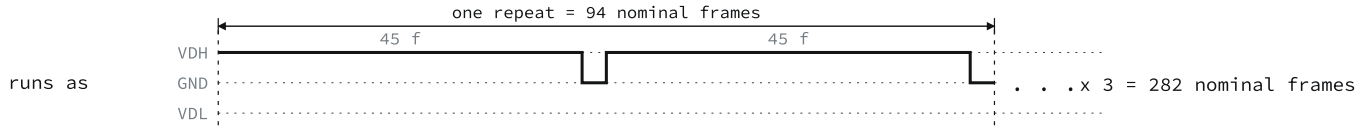
0xB0 payload, 400 B = 5 tables x 10 groups x 8 bytes

0 paper		1 ink		2 red		3 unused		4 VCOM	
g0	g1	g2	g3	g4	g5	g6	g7	g8	g9
level	on	off	ph2	on	off	ph2	repeat		
+0	+1	+2	+3	+4	+5	+6	+7		

level byte = 2 bits per phase, phase 0 in bits 7:6. 0 = GND, 1 = VDH, 2 = VDL, 3 = VDHR (reduced positive, red only).
offset of table t, group g, byte n = (t * 10 + g) * 8 + n

WORKED EXAMPLE table 1 (ink), group 1

	+0 level	+1 on	+2 off	+3 ph2	+4 on	+5 off	+6 ph2	+7 rep
bytes	0x44	45	1	1	45	1	1	3



A whole byte per count, so 45 fits where the 084 is capped at 15 by its nibbles. ph2 is always 1, so each half is drive, then rest.

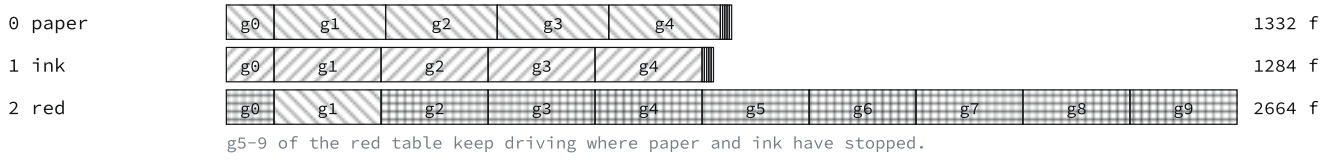
7 Complete table ALL 400 BYTES

TABLE	GROUP	BYTES	PHASE LEVELS	ON/OFF	REP	ROLE
0 paper	0	99 0A 0A 01 0A 0A 01 03	VDL VDH VDL VDH	10 / 10	3	split shake
0 paper	1–4	AA 2D 03 01 2D 03 01 03	VDL x 4	45 / 3	3	drive to paper, longer rest
0 paper	5–9	00 01 01 01 01 01 01 01	GND x 4	1 / 1	1	finished
1 ink	0	66 0A 0A 01 0A 0A 01 03	VDH VDL VDH VDL	10 / 10	3	split shake, opposite phase
1 ink	1–4	44 2D 01 01 2D 01 01 03	VDH GND VDH GND	45 / 1	3	drive to ink
1 ink	5–9	00 01 01 01 01 01 01 01	GND x 4	1 / 1	1	finished
2 red	0	EE 0A 0A 01 0A 0A 01 03	VDHR VDL VDHR VDL	10 / 10	3	split shake
2 red	1	AA 2D 01 01 2D 01 01 03	VDL x 4	45 / 1	3	clear before setting
2 red	2–9	FF 2D 01 01 2D 01 01 03	VDHR x 4	45 / 1	3	reduced positive rail
3 unused	0	00 0A 0A 01 0A 0A 01 03	GND x 4	10 / 10	3	unreachable, carries timing
3 unused	1–4	00 2D 01 01 2D 01 01 03	GND x 4	45 / 1	3	unreachable, carries timing
3 unused	5–9	00 01 01 01 01 01 01 01	GND x 4	1 / 1	1	—
4 VCOM	0–9	00 01 01 01 01 01 01 01	GND x 4	1 / 1	1	ground throughout

Hex: `0A`=10, `2D`=45, `03`=3, `01`=1. Tables 0 and 1 hold one half of the pixels each, so they get opposite drives and the shake is **split** — one level in every table would push the whole panel one way. Paper keeps the longer 45/3 rest: 45/1 everywhere cost brightness, L* 76.7 → 67.1. Measured: ink L* 15.1 (a* +1.88), paper L* 86.9, contrast 71.8, in **3460 ms** against 33 400 ms for the OTP waveform.

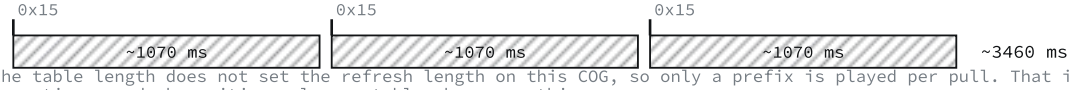
8 Why the table is pulled three times

WHAT THE TABLE ASKS FOR, IN NOMINAL FRAMES



g5-9 of the red table keep driving where paper and ink have stopped.

WHAT THE PANEL ACTUALLY RUNS



Not the same axis: the table length does not set the refresh length on this COG, so only a prefix is played per pull. That is why the dose is delivered three times, and why writing a longer table changes nothing.